

Modeling thermal ignition and pressurization in solid secondary explosives

B. F. Henson and L. Smilowitz

Chemistry Division, Los Alamos National Laboratory, Los Alamos, NM 87545

Work has been ongoing in our group to produce a global chemistry model of thermal ignition for organic secondary explosives valid over the entire temperature range of energetic response from thermal ignition by direct heating to detonation. We have made considerable progress recently, resulting in both the first broadly accurate global chemistry model of this type and the possible identification of a crucial component of the chemical mechanism governing solid thermal decomposition. We show that this crucial component is a general mechanism of crystalline decomposition based on the condensed phase thermodynamics of the explosive and therefore provides a universally applicable representation of the solid phase kinetics and a definition of the class of secondary explosives. Coupled with known gas phase reaction systems based on solid decomposition products the model successfully reproduces the rate of progress from crystalline solid to final products as a function of an initial thermal boundary condition. The model is similar in kind, but very different in detail from previous models produced by us and others. The model is based entirely on independently measured parameters for known processes in the chemistry of thermal decomposition and combustion in secondary explosives. We have applied the model in simple calculations of ignition time over the full temperature range of energetic response for 1,3,5,7-octahydro-1,3,5,7-tetranitro-tetrazocine (HMX), 1,3,5-triamino-2,4,6-trinitrobenzene (TATB) and 1,3-propanediol-2,2-bis[(nitrooxy)methyl]-tetranitrate (PETN), spanning 150 to 2500 C.

We show that these models reproduce times to ignition in the thermal ignition regime near the critical temperature for HMX, PETN and TATB. Combined with final temperatures that result from the coupled gas phase kinetics and final product chemistries it has also been shown to be consistent with the flame structure in the deflagration regime. With hypotheses regarding the thermal boundary condition in the shock regime this model also reproduces a temporal profile consistent with the detonation reaction zone of all three explosives.

In this talk we present a brief description and derivation of the parameters of the model for HMX, TATB and PETN and compare the predicted structure of thermal decomposition, ignition and detonation to experimental results for these explosives. By following the temporal trajectory of both the fluid (gas phase) density and temperature evolution during the later stages of reaction, and applying a simple JWL Equation of State approximation to obtain the pressure, it is possible to calculate the mechanical properties of the generated fluid. By taking derivatives of the product pressure as a function of time and density it is immediately apparent that highly non-ideal and incompressible states of the product fluid are potentially achieved during ignition by all three of

the secondary explosives examined here under thermal boundary conditions representative of thermal ignition or shock initiation.

These calculations afford some quantitative comparison with experiment for the coupling to mechanical boundaries and the resulting fracture and fragmentation. We will review measurements of internal pressure subsequent to thermal ignition and violent response in HMX and TATB based high explosive formulations from a number of laboratories. We will quantitatively compare the pressure evolution and more importantly the evolution of power contributed by reaction indicated by these calculations.